

```
In [1]: from sklearn.datasets import load_boston
import pandas as pd

boston_dataset = load_boston()

df = pd.DataFrame(boston_dataset.data, columns=boston_dataset.feature_names)
df['MEDV'] = boston_dataset.target

df.head(n=10)

In [2]: # If load_boston does not work then download the data and use this.
# Data : https://github.com/afnan47/sem8/blob/master/DL/1_boston_housing.csv
import pandas as pd
df = pd.read_csv("./1_boston_housing.csv")

In [3]: from sklearn.model_selection import train_test_split

X = df.loc[:, df.columns != 'MEDV']
y = df.loc[:, df.columns == 'MEDV']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state
=123)

In [4]: from sklearn.preprocessing import MinMaxScaler
mms = MinMaxScaler()
mms.fit(X_train)
X_train = mms.transform(X_train)
X_test = mms.transform(X_test)

In [5]: from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

model = Sequential()

model.add(Dense(128, input_shape=(13, ), activation='relu', name='dense_1'))
model.add(Dense(64, activation='relu', name='dense_2'))
model.add(Dense(1, activation='linear', name='dense_output'))
```

```
model.compile(optimizer='adam', loss='mse', metrics=['mae'])
model.summary()
```

Out:

Model: "sequential"

Layer (type)	Output Shape	Param #
=====		
dense_1 (Dense)	(None, 128)	1792
dense_2 (Dense)	(None, 64)	8256
dense_output (Dense)	(None, 1)	65
=====		
Total params: 10,113		
Trainable params: 10,113		
Non-trainable params: 0		
=====		

In [6]: history = model.fit(X_train, y_train, epochs=100, validation_split=0.05, verbose = 1)

Out:

```
Epoch 1/100
11/11 [=====] - 2s 57ms/step - loss: 1468.4281 - mae: 32.0408 - va
loss: 104.8761 - val_mae: 6.1343
Epoch 2/100
11/11 [=====] - 0s 9ms/step - loss: 291.9155 - mae: 13.9797 - val_
ss: 149.5561 - val_mae: 11.4423
Epoch 3/100
11/11 [=====] - 0s 7ms/step - loss: 151.9729 - mae: 10.3968 - val_
ss: 141.8171 - val_mae: 7.9483
Epoch 4/100
11/11 [=====] - 0s 8ms/step - loss: 87.9686 - mae: 7.0836 - val_lc
s: 96.4421 - val_mae: 8.3894
Epoch 5/100
11/11 [=====] - 0s 10ms/step - loss: 66.2962 - mae: 5.8493 - val_l
s: 79.1745 - val_mae: 5.6716
Epoch 6/100
11/11 [=====] - 0s 13ms/step - loss: 59.8653 - mae: 5.6803 - val_l
s: 81.4188 - val_mae: 5.8136
Epoch 7/100
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11/11 [=====] - 0s 7ms/step - loss: 57.8316 - mae: 5.3080 - val_loss: 78.1150 - val_mae: 6.1632
Epoch 8/100
11/11 [=====] - 0s 8ms/step - loss: 55.0569 - mae: 5.0945 - val_loss: 78.3047 - val_mae: 6.4755
Epoch 9/100
11/11 [=====] - 0s 4ms/step - loss: 56.2966 - mae: 5.2586 - val_loss: 78.7134 - val_mae: 6.3377
Epoch 10/100
11/11 [=====] - 0s 4ms/step - loss: 56.3019 - mae: 5.3718 - val_loss: 85.3209 - val_mae: 7.5740
Epoch 11/100
11/11 [=====] - 0s 4ms/step - loss: 65.1538 - mae: 6.0167 - val_loss: 82.1899 - val_mae: 5.8459
Epoch 12/100
11/11 [=====] - 0s 4ms/step - loss: 60.3453 - mae: 5.9333 - val_loss: 90.0579 - val_mae: 6.0229
Epoch 13/100
11/11 [=====] - 0s 8ms/step - loss: 59.1005 - mae: 5.7201 - val_loss: 93.0269 - val_mae: 6.0567
Epoch 14/100
11/11 [=====] - 0s 8ms/step - loss: 55.1477 - mae: 5.3231 - val_loss: 78.0659 - val_mae: 6.2727
Epoch 15/100
11/11 [=====] - 0s 11ms/step - loss: 49.6804 - mae: 4.9658 - val_loss: 78.2121 - val_mae: 6.2786
Epoch 16/100
11/11 [=====] - 0s 17ms/step - loss: 51.0314 - mae: 5.1608 - val_loss: 82.2059 - val_mae: 6.0243
Epoch 17/100
11/11 [=====] - 0s 12ms/step - loss: 50.8898 - mae: 5.0943 - val_loss: 79.5624 - val_mae: 6.7917
Epoch 18/100
11/11 [=====] - 0s 7ms/step - loss: 48.4888 - mae: 4.8631 - val_loss: 79.4000 - val_mae: 6.9262
Epoch 19/100
11/11 [=====] - 0s 8ms/step - loss: 47.7700 - mae: 4.8788 - val_loss: 78.2341 - val_mae: 6.6031
Epoch 20/100
11/11 [=====] - 0s 10ms/step - loss: 46.3437 - mae: 4.8531 - val_loss: 79.5777 - val_mae: 6.2096
Epoch 21/100
11/11 [=====] - 0s 15ms/step - loss: 46.8095 - mae: 4.8446 - val_loss: 79.9989 - val_mae: 6.2106
Epoch 22/100
11/11 [=====] - 0s 11ms/step - loss: 46.5442 - mae: 4.8957 - val_loss: 80.0767 - val_mae: 6.3410
```

Epoch 23/100
11/11 [=====] - 0s 5ms/step - loss: 46.3140 - mae: 4.7125 - val_loss: 79.3820 - val_mae: 6.3287

Epoch 24/100
11/11 [=====] - 0s 4ms/step - loss: 45.1415 - mae: 4.6794 - val_loss: 78.1009 - val_mae: 6.8739

Epoch 25/100
11/11 [=====] - 0s 5ms/step - loss: 46.8965 - mae: 4.8287 - val_loss: 77.1064 - val_mae: 6.9487

Epoch 26/100
11/11 [=====] - 0s 4ms/step - loss: 45.7256 - mae: 4.8219 - val_loss: 80.3456 - val_mae: 7.5219

Epoch 27/100
11/11 [=====] - 0s 3ms/step - loss: 45.4823 - mae: 4.8070 - val_loss: 86.8667 - val_mae: 7.9910

Epoch 28/100
11/11 [=====] - 0s 4ms/step - loss: 52.8585 - mae: 5.2916 - val_loss: 75.9413 - val_mae: 6.9428

Epoch 29/100
11/11 [=====] - 0s 4ms/step - loss: 43.7494 - mae: 4.7075 - val_loss: 75.8774 - val_mae: 6.8408

Epoch 30/100
11/11 [=====] - 0s 4ms/step - loss: 41.9384 - mae: 4.5150 - val_loss: 74.4587 - val_mae: 6.8040

Epoch 31/100
11/11 [=====] - 0s 4ms/step - loss: 42.0174 - mae: 4.6436 - val_loss: 72.5272 - val_mae: 6.0201

Epoch 32/100
11/11 [=====] - 0s 3ms/step - loss: 43.1446 - mae: 4.7250 - val_loss: 73.3913 - val_mae: 6.0527

Epoch 33/100
11/11 [=====] - 0s 5ms/step - loss: 42.5598 - mae: 4.5334 - val_loss: 70.9002 - val_mae: 6.6792

Epoch 34/100
11/11 [=====] - 0s 4ms/step - loss: 44.9650 - mae: 4.9500 - val_loss: 70.4410 - val_mae: 6.3134

Epoch 35/100
11/11 [=====] - 0s 4ms/step - loss: 44.2963 - mae: 4.7071 - val_loss: 76.1223 - val_mae: 5.9119

Epoch 36/100
11/11 [=====] - 0s 4ms/step - loss: 44.2435 - mae: 4.7664 - val_loss: 78.8820 - val_mae: 5.6082

Epoch 37/100
11/11 [=====] - 0s 4ms/step - loss: 39.0730 - mae: 4.4803 - val_loss: 69.7008 - val_mae: 6.2753

Epoch 38/100
11/11 [=====] - 0s 4ms/step - loss: 40.0643 - mae: 4.4467 - val_loss:

s: 70.4070 - val_mae: 5.9936
Epoch 39/100
11/11 [=====] - 0s 3ms/step - loss: 39.7373 - mae: 4.5365 - val_loss: 70.5524 - val_mae: 5.7211
Epoch 40/100
11/11 [=====] - 0s 4ms/step - loss: 40.1192 - mae: 4.3651 - val_loss: 68.0985 - val_mae: 6.1247
Epoch 41/100
11/11 [=====] - 0s 4ms/step - loss: 39.6499 - mae: 4.6153 - val_loss: 67.4054 - val_mae: 6.3753
Epoch 42/100
11/11 [=====] - 0s 4ms/step - loss: 41.4035 - mae: 4.6629 - val_loss: 67.4899 - val_mae: 5.8294
Epoch 43/100
11/11 [=====] - 0s 4ms/step - loss: 40.0755 - mae: 4.5632 - val_loss: 68.8983 - val_mae: 5.8097
Epoch 44/100
11/11 [=====] - 0s 4ms/step - loss: 39.9469 - mae: 4.5411 - val_loss: 67.7507 - val_mae: 6.1112
Epoch 45/100
11/11 [=====] - 0s 4ms/step - loss: 42.2766 - mae: 4.8104 - val_loss: 76.0111 - val_mae: 7.6458
Epoch 46/100
11/11 [=====] - 0s 4ms/step - loss: 41.7172 - mae: 4.8151 - val_loss: 67.7236 - val_mae: 5.6180
Epoch 47/100
11/11 [=====] - 0s 4ms/step - loss: 36.0112 - mae: 4.1907 - val_loss: 65.1950 - val_mae: 5.4492
Epoch 48/100
11/11 [=====] - 0s 5ms/step - loss: 34.7964 - mae: 4.2764 - val_loss: 73.0951 - val_mae: 5.5137
Epoch 49/100
11/11 [=====] - 0s 6ms/step - loss: 39.2266 - mae: 4.4734 - val_loss: 65.3500 - val_mae: 5.4734
Epoch 50/100
11/11 [=====] - 0s 4ms/step - loss: 36.9901 - mae: 4.4038 - val_loss: 65.8660 - val_mae: 6.4395
Epoch 51/100
11/11 [=====] - 0s 4ms/step - loss: 36.3954 - mae: 4.3774 - val_loss: 66.5157 - val_mae: 5.5489
Epoch 52/100
11/11 [=====] - 0s 3ms/step - loss: 35.5653 - mae: 4.3058 - val_loss: 65.4602 - val_mae: 5.6216
Epoch 53/100
11/11 [=====] - 0s 4ms/step - loss: 34.9797 - mae: 4.3144 - val_loss: 71.0284 - val_mae: 5.6592
Epoch 54/100

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11/11 [=====] - 0s 4ms/step - loss: 34.9526 - mae: 4.3036 - val_loss: 79.4032 - val_mae: 5.6828
Epoch 55/100
11/11 [=====] - 0s 4ms/step - loss: 35.1356 - mae: 4.1692 - val_loss: 63.7029 - val_mae: 6.1306
Epoch 56/100
11/11 [=====] - 0s 4ms/step - loss: 35.1635 - mae: 4.3432 - val_loss: 64.8605 - val_mae: 5.7787
Epoch 57/100
11/11 [=====] - 0s 4ms/step - loss: 34.1723 - mae: 4.2143 - val_loss: 73.1095 - val_mae: 5.7776
Epoch 58/100
11/11 [=====] - 0s 4ms/step - loss: 34.2169 - mae: 4.0968 - val_loss: 75.7989 - val_mae: 6.0117
Epoch 59/100
11/11 [=====] - 0s 6ms/step - loss: 37.3029 - mae: 4.4993 - val_loss: 69.9205 - val_mae: 5.8598
Epoch 60/100
11/11 [=====] - 0s 5ms/step - loss: 36.1463 - mae: 4.3774 - val_loss: 93.1340 - val_mae: 6.6789
Epoch 61/100
11/11 [=====] - 0s 3ms/step - loss: 35.3942 - mae: 4.1797 - val_loss: 69.6372 - val_mae: 6.9278
Epoch 62/100
11/11 [=====] - 0s 3ms/step - loss: 44.9558 - mae: 5.2795 - val_loss: 74.5363 - val_mae: 5.8979
Epoch 63/100
11/11 [=====] - 0s 4ms/step - loss: 40.2197 - mae: 4.6395 - val_loss: 67.6870 - val_mae: 5.7321
Epoch 64/100
11/11 [=====] - 0s 4ms/step - loss: 35.6249 - mae: 4.4874 - val_loss: 63.1919 - val_mae: 6.0185
Epoch 65/100
11/11 [=====] - 0s 5ms/step - loss: 33.5510 - mae: 4.1380 - val_loss: 75.4116 - val_mae: 5.7737
Epoch 66/100
11/11 [=====] - 0s 4ms/step - loss: 33.3171 - mae: 4.3020 - val_loss: 65.2549 - val_mae: 5.6731
Epoch 67/100
11/11 [=====] - 0s 4ms/step - loss: 30.4981 - mae: 3.9988 - val_loss: 65.9374 - val_mae: 6.0587
Epoch 68/100
11/11 [=====] - 0s 4ms/step - loss: 31.4584 - mae: 4.0822 - val_loss: 70.9584 - val_mae: 5.9610
Epoch 69/100
11/11 [=====] - 0s 4ms/step - loss: 32.0598 - mae: 4.0980 - val_loss: 65.5629 - val_mae: 6.0606
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Epoch 70/100
11/11 [=====] - 0s 4ms/step - loss: 30.0251 - mae: 3.9024 - val_loss: 66.6761 - val_mae: 5.7830
Epoch 71/100
11/11 [=====] - 0s 4ms/step - loss: 30.8111 - mae: 4.1027 - val_loss: 73.0866 - val_mae: 5.8113
Epoch 72/100
11/11 [=====] - 0s 3ms/step - loss: 32.4922 - mae: 4.1465 - val_loss: 65.9663 - val_mae: 5.8200
Epoch 73/100
11/11 [=====] - 0s 6ms/step - loss: 29.3212 - mae: 4.0037 - val_loss: 68.6681 - val_mae: 5.7642
Epoch 74/100
11/11 [=====] - 0s 4ms/step - loss: 28.9626 - mae: 3.9230 - val_loss: 70.9759 - val_mae: 5.7254
Epoch 75/100
11/11 [=====] - 0s 4ms/step - loss: 27.2850 - mae: 3.7621 - val_loss: 65.1627 - val_mae: 5.7805
Epoch 76/100
11/11 [=====] - 0s 4ms/step - loss: 29.2583 - mae: 3.9348 - val_loss: 63.4590 - val_mae: 5.9799
Epoch 77/100
11/11 [=====] - 0s 6ms/step - loss: 31.6208 - mae: 4.1665 - val_loss: 71.6206 - val_mae: 5.7315
Epoch 78/100
11/11 [=====] - 0s 5ms/step - loss: 29.2416 - mae: 3.9052 - val_loss: 63.1969 - val_mae: 5.9108
Epoch 79/100
11/11 [=====] - 0s 4ms/step - loss: 28.9797 - mae: 3.9858 - val_loss: 79.6432 - val_mae: 6.1192
Epoch 80/100
11/11 [=====] - 0s 4ms/step - loss: 30.1288 - mae: 4.0720 - val_loss: 64.5131 - val_mae: 5.7942
Epoch 81/100
11/11 [=====] - 0s 3ms/step - loss: 27.6364 - mae: 3.8092 - val_loss: 65.6751 - val_mae: 5.5210
Epoch 82/100
11/11 [=====] - 0s 4ms/step - loss: 28.2396 - mae: 3.9231 - val_loss: 91.4888 - val_mae: 6.3524
Epoch 83/100
11/11 [=====] - 0s 4ms/step - loss: 40.4750 - mae: 4.8559 - val_loss: 60.2526 - val_mae: 5.8334
Epoch 84/100
11/11 [=====] - 0s 4ms/step - loss: 37.2588 - mae: 4.8038 - val_loss: 68.2171 - val_mae: 5.4533
Epoch 85/100
11/11 [=====] - 0s 4ms/step - loss: 30.8255 - mae: 3.9874 - val_loss:
```

s: 67.0612 - val_mae: 5.5345
Epoch 86/100
11/11 [=====] - 0s 4ms/step - loss: 27.5965 - mae: 3.8464 - val_loss: 63.5904 - val_mae: 5.8538
Epoch 87/100
11/11 [=====] - 0s 4ms/step - loss: 27.9007 - mae: 3.9001 - val_loss: 79.7395 - val_mae: 5.8449
Epoch 88/100
11/11 [=====] - 0s 3ms/step - loss: 29.3778 - mae: 4.0870 - val_loss: 64.1248 - val_mae: 6.0386
Epoch 89/100
11/11 [=====] - 0s 4ms/step - loss: 28.6313 - mae: 3.9004 - val_loss: 64.7652 - val_mae: 5.6717
Epoch 90/100
11/11 [=====] - 0s 4ms/step - loss: 26.2589 - mae: 3.7857 - val_loss: 73.6284 - val_mae: 5.6817
Epoch 91/100
11/11 [=====] - 0s 4ms/step - loss: 25.5219 - mae: 3.6840 - val_loss: 61.2618 - val_mae: 6.2671
Epoch 92/100
11/11 [=====] - 0s 4ms/step - loss: 37.5101 - mae: 4.7214 - val_loss: 70.1923 - val_mae: 5.4056
Epoch 93/100
11/11 [=====] - 0s 4ms/step - loss: 30.1798 - mae: 4.1516 - val_loss: 59.7963 - val_mae: 6.0707
Epoch 94/100
11/11 [=====] - 0s 4ms/step - loss: 29.6977 - mae: 4.0330 - val_loss: 57.3263 - val_mae: 5.6396
Epoch 95/100
11/11 [=====] - 0s 4ms/step - loss: 27.2931 - mae: 3.9198 - val_loss: 77.2184 - val_mae: 5.6756
Epoch 96/100
11/11 [=====] - 0s 4ms/step - loss: 30.7540 - mae: 4.0007 - val_loss: 63.6341 - val_mae: 6.7858
Epoch 97/100
11/11 [=====] - 0s 5ms/step - loss: 30.7228 - mae: 4.1860 - val_loss: 69.4769 - val_mae: 5.6059
Epoch 98/100
11/11 [=====] - 0s 4ms/step - loss: 26.2480 - mae: 3.8078 - val_loss: 65.4580 - val_mae: 5.5055
Epoch 99/100
11/11 [=====] - 0s 4ms/step - loss: 25.7807 - mae: 3.6395 - val_loss: 67.9779 - val_mae: 5.6516
Epoch 100/100
11/11 [=====] - 0s 4ms/step - loss: 25.8939 - mae: 3.7184 - val_loss: 63.8970 - val_mae: 5.7290


```
In [7]: mse_nn, mae_nn = model.evaluate(X_test, y_test)

print('Mean squared error on test data: ', mse_nn)
print('Mean absolute error on test data: ', mae_nn)
```

Out:

```
5/5 [=====] - 0s 2ms/step - loss: 25.9594 - mae: 3.8497
Mean squared error on test data: 25.959447860717773
Mean absolute error on test data: 3.8496510982513428
```